

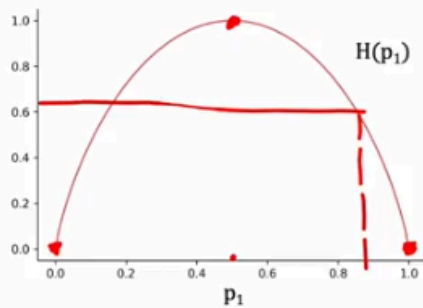
Day 26. Decision tree learning

Decision tree learning

Measuring purity

Entropy as a measure of impurity

p_1 = fraction of examples that are cats



$$p_1 = 3/6 \quad H(p_1) = 1$$

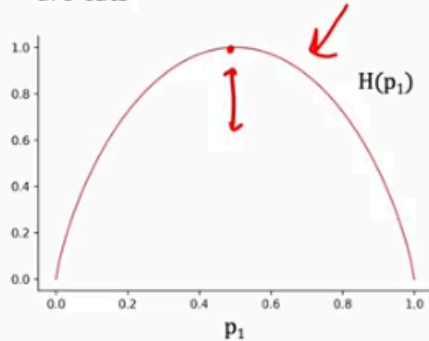
$$p_1 = 5/6 \quad H(p_1) = 0.65$$

$$p_1 = 6/6$$

we use entropy here to represent the impurity, the higher the entropy, the higher impurity.
the entropy function is as below

Entropy as a measure of impurity

p_1 = fraction of examples that are cats



$$p_0 = 1 - p_1$$

$$\begin{aligned} H(p_1) &= -p_1 \log_2(p_1) - p_0 \log_2(p_0) \\ &= -p_1 \log_2(p_1) - (1 - p_1) \log_2(1 - p_1) \end{aligned}$$

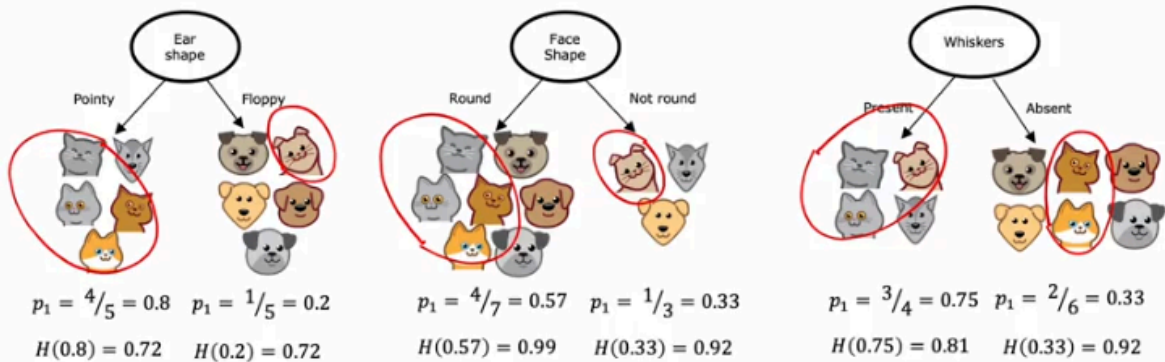
Note: " $0 \log(0)$ " = 0

Choosing a split: Information Gain

when building a decision tree, the way we'll decide what feature to split on at a node will be based on what choice of feature reduces entropy the most.

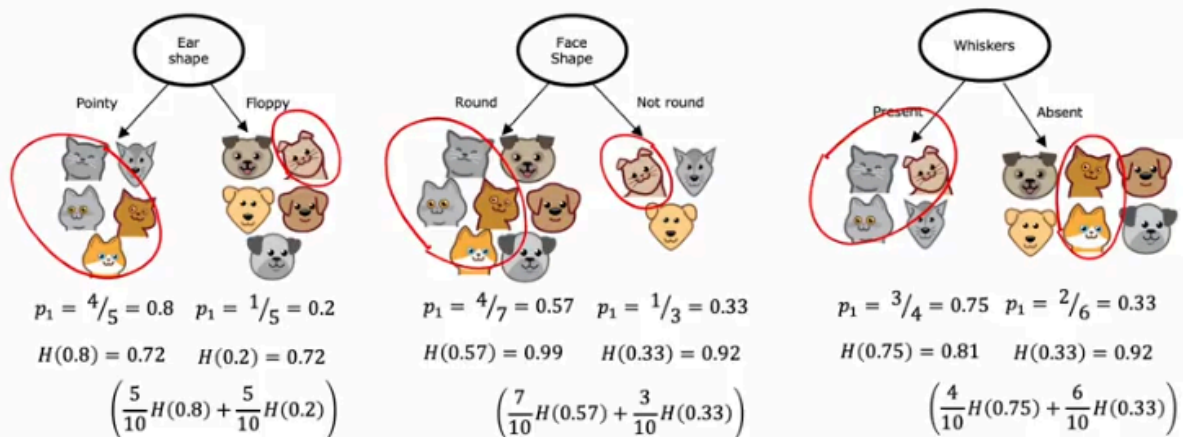
The reduction of the entropy is called information gain.

Choosing a split



however, we don't just directly compare the entropies of the splits, since groups with less objects but with high entropy are surely better than the high-entropy groups with more objects.

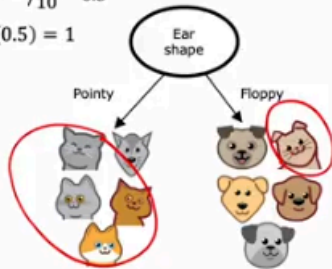
Choosing a split



Choosing a split

$$p_1 = 5/10 = 0.5$$

$$H(0.5) = 1$$



$$p_1 = 4/5 = 0.8 \quad p_1 = 1/5 = 0.2$$

$$H(0.8) = 0.72 \quad H(0.2) = 0.72$$

$$H(0.5) - \left(\frac{5}{10} H(0.8) + \frac{5}{10} H(0.2) \right)$$

$$= 0.28$$

$$H(0.5) = 1$$



$$p_1 = 4/7 = 0.57 \quad p_1 = 1/3 = 0.33$$

$$H(0.57) = 0.99 \quad H(0.33) = 0.92$$

$$H(0.5) - \left(\frac{7}{10} H(0.57) + \frac{3}{10} H(0.33) \right)$$

$$= 0.03$$

$$H(0.5) = 1$$



$$p_1 = 3/4 = 0.75 \quad p_1 = 2/6 = 0.33$$

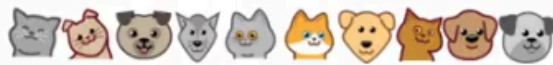
$$H(0.75) = 0.81 \quad H(0.33) = 0.92$$

$$H(0.5) - \left(\frac{4}{10} H(0.75) + \frac{6}{10} H(0.33) \right)$$

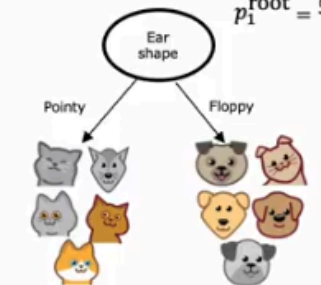
$$= 0.12$$

Information gain

Information Gain



$$p_1^{\text{root}} = 5/10 = 0.5$$



$$p_1^{\text{left}} = 4/5 \quad p_1^{\text{right}} = 1/5$$

$$w^{\text{left}} = 5/10 \quad w^{\text{right}} = 5/10$$

Information gain

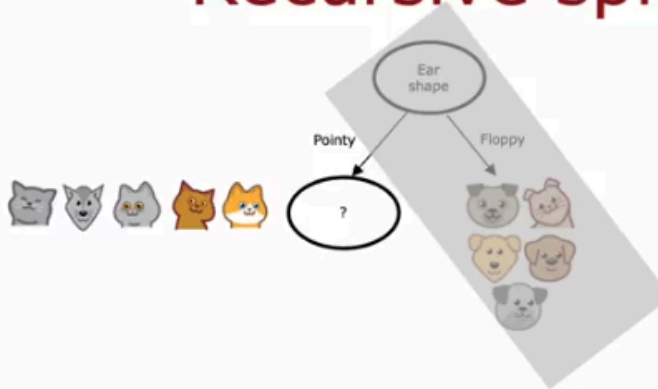
$$= H(p_1^{\text{root}}) - \left(w^{\text{left}} H(p_1^{\text{left}}) + w^{\text{right}} H(p_1^{\text{right}}) \right)$$

Putting it together(build a decision tree)

Decision Tree Learning

- Start with all examples at the root node
- Calculate information gain for all possible features, and pick the one with the highest information gain
- Split dataset according to selected feature, and create left and right branches of the tree
- Keep repeating splitting process until stopping criteria is met:
 - When a node is 100% one class
 - When splitting a node will result in the tree exceeding a maximum depth
 - Information gain from additional splits is less than threshold
 - When number of examples in a node is below a threshold

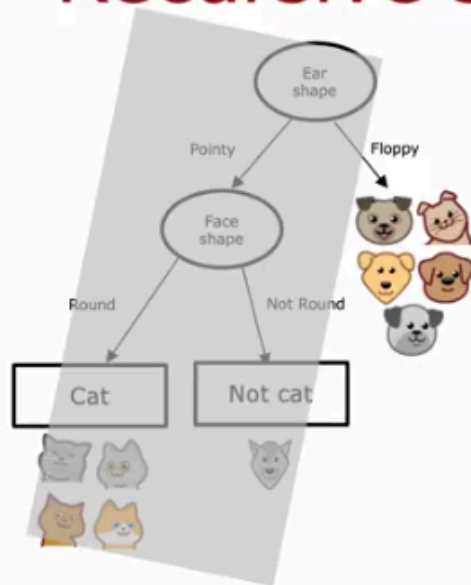
Recursive splitting



Recursive splitting



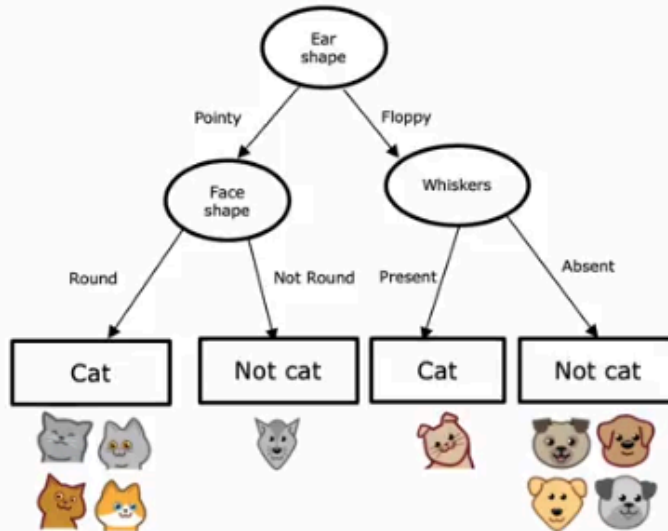
Recursive splitting



Recursive splitting



Recursive splitting



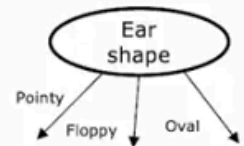
Using one-hot encoding of categorical features

above we only use features that have only two possibilities. now we are going to use one-hot encoding to model address features with multiple results.

Features with three possible values

	Ear shape (x_1)	Face shape (x_2)	Whiskers (x_3)	Cat (y)
	Pointy	Round	Present	1
	Oval	Not round	Present	1
	Oval	Round	Absent	0
	Pointy	Not round	Present	0
	Oval	Round	Present	1
	Pointy	Round	Absent	1
	Floppy	Not round	Absent	0
	Oval	Round	Absent	1
	Floppy	Round	Absent	0
	Floppy	Round	Absent	0

3 possible values



One hot encoding

Ear shape	Pointy ears	Floppy ears	Oval ears	Face shape	Whiskers	Cat	
	Pointy	1	0	0	Round	Present	1
	Oval	0	0	1	Not round	Present	1
	Oval	0	0	1	Round	Absent	0
	Pointy	1	0	0	Not round	Present	0
	Oval	0	0	1	Round	Present	1
	Pointy	1	0	0	Round	Absent	1
	Floppy	0	1	0	Not round	Absent	0
	Oval	0	0	1	Round	Absent	1
	Floppy	0	1	0	Round	Absent	0
	Floppy	0	1	0	Round	Absent	0

One hot encoding

If a categorical feature can take on k values, create k binary features (0 or 1 valued).

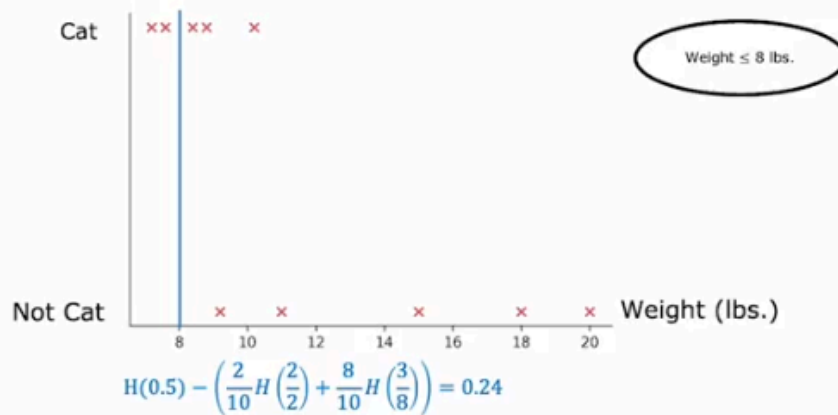
One hot encoding and neural networks

	Pointy ears	Floppy ears	Round ears	Face shape	Whiskers	Cat
	1	0	0	Round 1	Present 1	1
	0	0	1	Not round 0	Present 1	1
	0	0	1	Round 1	Absent 0	0
	1	0	0	Not round 0	Present 1	0
	0	0	1	Round 1	Present 1	1
	1	0	0	Round 1	Absent 0	1
	0	1	0	Not round 0	Absent 0	1
	0	0	1	Round 1	Absent 0	1
	0	1	0	Round 1	Absent 0	1
	0	1	0	Round 1	Absent 0	1

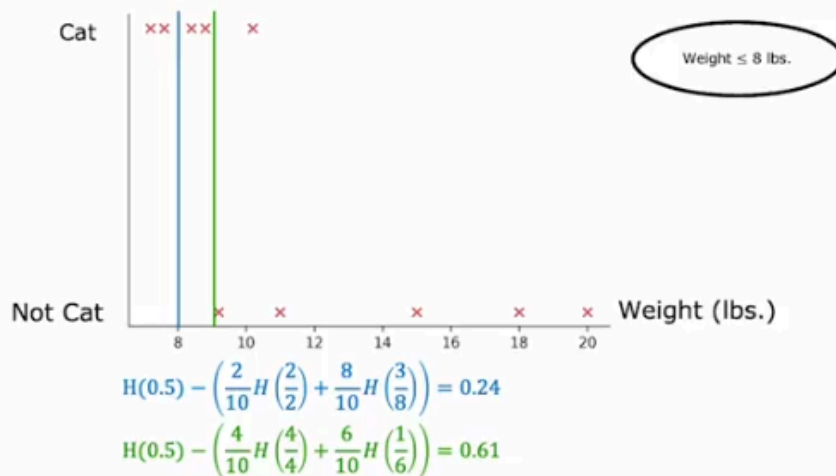
let's take a step further, how to deal with continuous features?

Continuous valued features

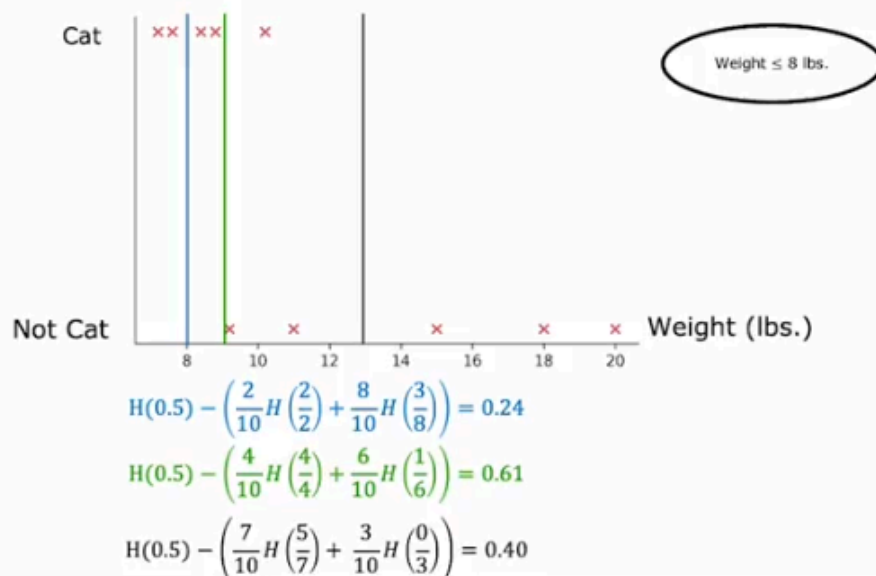
Splitting on a continuous variable



Splitting on a continuous variable



Splitting on a continuous variable



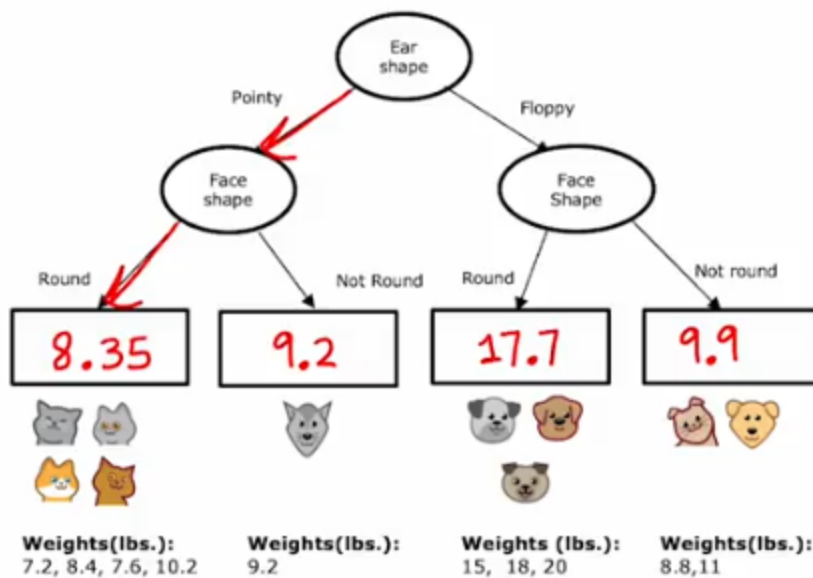
Regression trees

Regression with Decision Trees: Predicting a number

	Ear shape	Face shape	Whiskers	Weight (lbs.)
	Pointy	Round	Present	7.2
	Floppy	Not round	Present	8.8
	Floppy	Round	Absent	15
	Pointy	Not round	Present	9.2
	Pointy	Round	Present	8.4
	Pointy	Round	Absent	7.6
	Floppy	Not round	Absent	11
	Pointy	Round	Absent	10.2
	Floppy	Round	Absent	18
	Floppy	Round	Absent	20

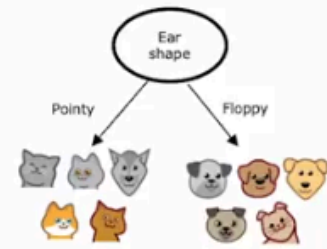
X y

Regression with Decision Trees



instead of choosing splits according to entropy, we are going to use weighted variance here.

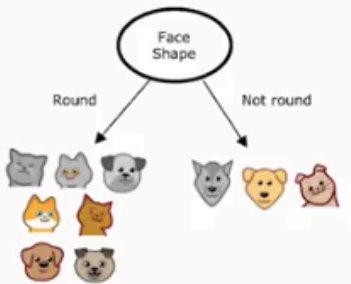
Choosing a split



Weights: 7.2, 9.2, 8.4, 7.6, 10.2
 Variance: 1.47
 $w^{\text{left}} = \frac{5}{10}$

Weights: 8.8, 15, 11, 18, 20
 Variance: 21.87
 $w^{\text{right}} = \frac{5}{10}$

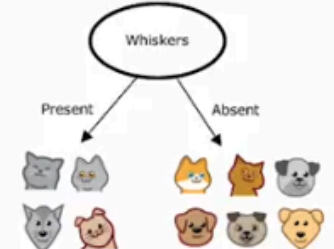
$$\left(\frac{5}{10} * 1.47 + \frac{5}{10} * 21.87\right)$$



Weights: 7.2, 15, 8.4, 7.6, 10.2, 18, 20
 Variance: 27.80
 $w^{\text{left}} = \frac{7}{10}$

Weights: 8.8, 9.2, 11
 Variance: 1.37
 $w^{\text{right}} = \frac{3}{10}$

$$\left(\frac{7}{10} * 27.80 + \frac{3}{10} * 1.37\right)$$

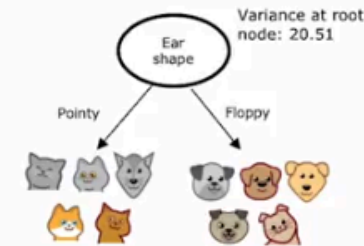


Weights: 7.2, 8.8, 9.2, 8.4
 Variance: 0.75
 $w^{\text{left}} = \frac{4}{10}$

Weights: 15, 7.6, 11, 10.2, 18, 20
 Variance: 23.32
 $w^{\text{right}} = \frac{6}{10}$

$$\left(\frac{4}{10} * 0.75 + \frac{6}{10} * 23.32\right)$$

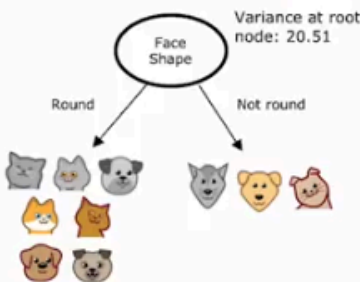
Choosing a split



Weights: 7.2, 9.2, 8.4, 7.6, 10.2
 Variance: 1.47
 $w^{\text{left}} = \frac{5}{10}$

Weights: 8.8, 15, 11, 18, 20
 Variance: 21.87
 $w^{\text{right}} = \frac{5}{10}$

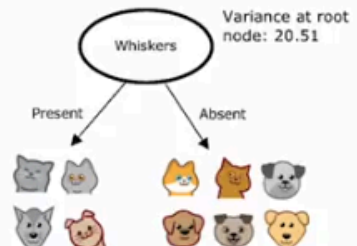
$$20.51 - \left(\frac{5}{10} * 1.47 + \frac{5}{10} * 21.87\right) = 8.84$$



Weights: 7.2, 15, 8.4, 7.6, 10.2, 18, 20
 Variance: 27.80
 $w^{\text{left}} = \frac{7}{10}$

Weights: 8.8, 9.2, 11
 Variance: 1.37
 $w^{\text{right}} = \frac{3}{10}$

$$20.51 - \left(\frac{7}{10} * 27.80 + \frac{3}{10} * 1.37\right) = 0.64$$



Weights: 7.2, 8.8, 9.2, 8.4
 Variance: 0.75
 $w^{\text{left}} = \frac{4}{10}$

Weights: 15, 7.6, 11, 10.2, 18, 20
 Variance: 23.32
 $w^{\text{right}} = \frac{6}{10}$

$$20.51 - \left(\frac{4}{10} * 0.75 + \frac{6}{10} * 23.32\right) = 6.22$$